MILITARY SPECIFICATION

GAGE, HEIGHT, ELECTRONIC INDICATING

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers electronically amplified height gages.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Method

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANS C39.1-1964	Electrical Indicating Instruments, Requirements for
ANS C39.5-1964	Electrical Measuring and Controlling Instrumentation, Safety Requirements for
ANS B46.1-1962	Surface Texture

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(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN TRUCKING ASSOCIATION, INC (ATA)

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W Jackson Blvd., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), the contractor shall furnish a complete gage assembly for preproduction inspection in accordance with Section 4. The sample may be a preproduction item or a production item which meets the requirements of this specification. In either case, the sample shall be identical with the proposed items and shall be made or shall have been made by the methods that are to be used or were used in production of items to fulfill the terms of the contract. The preproduction sample may be shipped as part of the contract. Approval of a preproduction sample shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Safety. The gage assembly shall have adequate safety provisions to protect the operator from injury and electrical shock.

3.3 Design. The gage assembly shall be of the manufacturer's latest design that has been commercially proven through usage and shall essentially consist of an electronic amplifier, gaging head, universal clamp, upright column and support base, gaging head crossbar, electrical wire and connections.

3.3.1 Interchangeability. All parts shall be manufactured to such standards as will permit ease of adjustment and replacement without modification of parts.

3.3.2 Material. All parts used in fabrication of the gage assembly shall be new and of the quality necessary to produce a gage that will meet the requirements specified herein.

3.4 Construction. The gage assembly shall be complete so that when installed and connected to the specified source of power it can be used for any operation for which it was designed. Construction shall be free from any characteristics or defects that will prevent the gage assembly from passing any of the examinations and tests in Section 4.

3.4.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of the assembly.

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3.4.2 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.3 Surfaces. All surfaces of all castings, forging, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.4.4 Painting. Unless otherwise specified (see 6.2), the gage assembly shall be painted in accordance with the manufacturer's standard commercial practice.

3.5 Components.

3.5.1 Electronic amplifier. The amplifier shall be a transistorized unit constructed of material currently approved by the electronic gaging industry. Unless otherwise specified (see 6.2), the amplifier shall have not less than two distinct measurement ranges with related read-out scales with division values and measurement ranges as specified in Table I. The measurement ranges may be direct, multi-scale read-out, with a scale selector knob, or a single scale with a range selector knob. The amplifier shall be equipped with an off-on switch, provisions for calibrating, provisions for zero setting, a power supply and a recorder plug-in socket and a carrying handle. When specified (see 6.2), the equipment shall have the capability of indicating signals from two gage heads, either singly, in additive combinations, or in differential combinations. Specific features not included herein shall be in accordance with standards ANS C39.1-1964 and ANS C39.5-1964.

3.5.2 Range and scale divisions. In regards to Table I, amplifiers may have intermediate scale ranges between the "high" and the "low" amplifier scales stated, but its overall ranges shall encompass the stated high or less to the stated low or greater.

TABLE I

AMPLIFIER SCALES	RANGE	MEASUREMENT PER DIVISION
High	± .0002	.000010
Low	± .002	.000100

3.5.3 Gaging head. The gaging head shall be the reversible type for over and under gaging operations with gage contact actuating force of not more than 10 grams for both over and under gaging. The gaging head shall have provisions for being attached onto an adjustable crossbar. Unless otherwise specified (see 6.2), the gaging head dimensions shall be not more than 2-3/4 inches long by 1-7/16 inches high by 7/8 inches wide.

3.5.3.1 Gage probes. Gage probes shall be designed to be readily interchangeable, and shall be so mounted to the gage heads so that they will be able to withstand displacement in any direction without causing damage to the gaging

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head or to the probe.

3.5.4 Universal clamp. The clamp shall be a snug, sliding fit on the column and gaging head support crossbar. The clamp shall have provisions for simultaneous and independent adjustments on the column, crossbar and swiveling features. Each adjustment feature shall have provisions for being secured in place.

3.5.5 Column and base. The column shall be attached to the base to form an integral assembly and shall be of sufficient rigidity to insure repetitive measurements for the full measuring range. Unless otherwise specified (see 6.2), the column shall be not less than 24 inches high with surface finish not rougher than 16 microinches in accordance with ANSI B46.1. The base shall be iron, iron alloy, or semi steel and shall have three point base contact. Unless otherwise specified (see 6.2), the respective length, width, and height of the base shall be not less than 6-1/4 X 3-3/4 X 2-1/4 inches.

3.5.6 Fine height adjustment. A fine height adjustment feature of a range of not less than .030 inches shall be incorporated in the gage assembly.

3.5.7 Gaging head crossbar. Unless otherwise specified (see 6.2), a crossbar shall be provided that will allow height measurements to be made for a horizontal distance of not less than 10 inches away from the centerline of the vertical column. The cross bar shall be of sufficient rigidity to insure repetitive measurements when extended to the maximum position. Surface finish shall be not rougher than 16 microinches and shall be a snug slip fit in the universal clamp. One end of the crossbar shall have provisions for mounting and adjusting the gaging head for measuring both vertical and horizontal surfaces.

3.5.8 Direct current power requirements. If there are any requirements of the system for direct current, the system shall include provisions for converting alternating current into direct current.

3.6 Electrical equipment.

3.6.1 Power. Unless otherwise specified (see 6.2), the equipment shall be operable from an input power of 115 to 120 volts, 60 hertz, single phase, alternating current.

3.6.2 Power supply cord. The cord shall be 58" minimum length, three-wire, oil resistant, rubber insulating cord with voltage and current capacity not less than the amplifier rating required. One wire shall serve as an effective ground to the amplifier and the opposite end of the cord shall terminate in a polarized three-wire grounding plug.

3.7 Nameplate. A corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each amplifier.

Nomenclature
Manufacturer's Model Number
Manufacturer's Serial Number
Special Characteristics (Voltage, etc)
Stock Number
Manufacturer's Name

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Contract Number
Date of Manufacture
US

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable to the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Preproduction inspection.

b. Quality conformance inspection.

4.2.1 Preproduction inspection. Preproduction inspection shall be applied to the preproduction sample submitted in accordance with 3.1. Failure of the preproduction sample to pass the examination or one or more preproduction tests shall be cause for rejection. Preproduction inspection shall consist of the examination in 4.3 and all tests in 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Failure of the gage assembly to pass the examination or one or more tests shall be cause for rejection. Quality conformance inspection shall consist of the following:

a. Examination (4.3)

b. Operational test (4.4.1).

c. Inspection of preservation, packaging, packing and marking for shipment and storage (4.5).

4.3 Examination. The gage assembly shall be examined for design, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.4 Tests.

4.4.1 Operational test. The gage assembly shall be operated for not less than 30 minutes. Proper operation of all controls, adjusting mechanism and other accessories shall be inspected during the trial period.

4.4.2 Functional test. Amplifier meter response shall be not more than 1/4 second without overshoot. Zero point shall coincide on multi scale amplifiers. The gaging assembly shall be tested for repetitive accuracy in the following manner: The gaging head shall be positioned at three distinct

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locations on the column, the extreme low, the extreme high, and center. Not less than three repetitive measurements shall be made at each location on the gage head using the smallest incremental scale range required. Reverse the gaging contact point and repeat the above sequence. Measurement deviation at any gage head location shall be not more than 1/2 scale division.

4.5 Inspection of preservation, packaging, packing, and marking for shipment and storage. Preservation, packaging, packing, and marking for shipment and storage shall be inspected to determine conformance with the requirements specified in Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Levels A, B, or C as specified (see 6.2).

5.1.1 Levels A and B. Gage, consisting of the following components: electronic amplifier, gaging head, universal clamp, column and base, and gaging head crossbar, shall be preserved and packaged in accordance with methods of MIL-P-116 as follows:

COMPONENT	METHOD
Electronic Amplifier	II
Gaging Head	II
Universal Clamp	I P-6 Preservative
Column and Base	I P-6 Preservative
Gaging Head Crossbar	I P-6 Preservative

5.1.2 Level C. Gage consisting of components as listed in para 5.1.1 shall be packaged in accordance with the manufacturer's standard commercial practice.

5.2 Packing. Packing shall be levels A, B or C as specified (see 6.2).

5.2.1 Level A. Gage consisting of components preserved and packaged as indicated in para 5.1.1 shall be overpacked into an exterior, snug fitting, overseas, wood cleated plywood box, PPP-B-601, style optional. Exterior container insofar as practicable, shall be of a uniform size and minimum cube consistent with the protection required. Gross weight shall not exceed that as specified in the referenced container specification. Container shall be closed and sealed in accordance with the container specification and appendix thereto.

5.2.1.1 Packing for level A. Place package containing column and base and the package containing the gaging head crossbar into the bottom of the container. Block and brace to prevent movement in accordance with MIL-STD-1186, using wood blocking and bracing. Cover with 1/4" plywood cut to interior dimensions (length and width) of container and secure in place as a divider using 2" X 2" wood cleats nailed along both sides and ends (interior) of the container. The remaining packaged items, consisting of electronic amplifier, gaging head and universal clamps, shall be packed in the top section of the container. Fill all voids with cushioning material, PPP-C-843.

5.2.2 Level B. Gage, consisting of components, preserved and packaged as indicated in para 5.1.1 shall be packed the same as level A except that the container shall be domestic type.

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5.2.3 Level C. Gage, consisting of components, preserved and packaged as indicated in para 5.1.1 or 5.1.2 shall be packaged to assure carriers acceptance and safe delivery to destination at the lowest rates. Containers shall meet Uniform Freight Classification Rules or National Motor Freight Classification Rules, or regulations of other common carriers, as applicable to the mode of transportation.

5.3 Marking. In addition to any special markings as required by the contract or order (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This equipment will be used for height measurements where a high degree of accuracy is required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Preproduction inspection, if required (see 3.1 and 4.2.1).

c. Painting, if different (see 3.4.4).

d. Amplifier measurement indication ranges, if different (see 3.5.1).

e. Twin measurement head capability of amplifier indicator, if required (see 3.5.1).

f. Dimensions of gaging head, if different (see 3.5.3).

g. Column height and roughness, if different (see 3.5.5).

h. Base dimensions, if different (see 3.5.5)

i. Horizontal measurement reach, if different (see 3.5.7).

j. Input power requirements, if different (see 3.6.1).

k. Selection of applicable levels of preservation and packing (see 5.1 and 5.2)

l. Marking required (see 5.3).

Custodians:
Army - GL
Navy - YD
Air Force - 84

User activities:
Navy - MC, OS

Preparing activity:
Air Force - 84

Project number:
5210-0213

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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